

Feasibility Study on Vermicomposting at Northern Arizona University

I. Introduction

Vermicomposting offers a biologically efficient, low energy method for diverting food waste from landfills using Red Wiggler (*Eisenia fetida*) worms. At Northern Arizona University, interest in vermicomposting has grown alongside broader composting programs—particularly as the university seeks to reduce reliance on landfills, extend the lifespan of the Cinder Lake Landfill, and cut greenhouse gas emissions. This feasibility study evaluates whether expanded vermicomposting operations can become a long term, institutionally supported component of NAU's sustainability infrastructure. The current composting methods on campus are evaluated and compared with vermicomposting.

A Vermicomposting Living Lab was established at the Compost Microbiome Lab located at the Environmental Engineering Field Station on NAU's south campus to process organic waste from campus dining halls. Red Wiggler (*Eisenia fetida*) worms break down organic materials such as vegetable scraps and paper, converting them into nutrient-rich soil. Funded by the NAU Green Fund, the lab serves as both a research site and a hands-on learning space for NAU students and community members interested in composting systems, soil health, and sustainable waste management.

By diverting organic material from landfills, the lab reduces methane emissions and supports NAU's Climate Action Plan. This lab provides an additional on-campus composting system that processes organic waste and produces an ecologically and economically beneficial end product, castings. Worm castings (worm poop) provide slow release nutrients, improve water retention, enhance soil structure, and promote healthy microbial communities.

Currently, compost and casting samples are being collected for sequencing based analyses to track changes in microbial communities and their functional potential, thereby improving composting system efficiency. This accompanying feasibility study explores the potential to expand campus wide vermicomposting across student residences, offices, and larger scale applications capable of processing most of the university's food waste. Through campus collaborations and workshops, we hope to raise awareness about vermicomposting and continue to support on campus sustainability.

II. History & Current Status of Composting and Vermicomposting at NAU

A. NAU's Composting Infrastructure: Historical Overview

NAU's composting program has evolved significantly over the past 15 years, with major milestones documented in the NAU sustainability outline. In 2012, NAU implemented a composting system with VeloComposting. This was a student powered pilot program to initiate large scale composting on campus. Students would use their bikes with trailers to pick up used food scraps and coffee grounds from several dining halls on campus. They would then bring the food waste to the South Campus for composting (Northern Arizona University, n.d.).

In response to NAU's pilot composting project creation, a former forestry student conducted a year and a half long study to develop the most effective large scale composting on campus. During the investigation, the student found that the amount of organic material deposited into Flagstaff's Cinder Lake Landfill accounted for 28% of the city's waste stream. The amount of food waste deposited in the landfill each year that could be composted was found to be 60,000 tons. In response to the student's efforts, NAU created a composting program to reduce overall waste sent to the landfill. During an 18-month study, it was shown that NAU composted ~10,000 pounds per week. This means NAU is projected to produce a maximum of ~520,000 pounds per year (Almehmadi et al., 2019a).

In 2013, the program became an official part of the University's operations, with NAU taking over the composting initiative. Moving into 2015-2017, NAU Campus Dining upgraded by implementing a new post-consumer waste composting processor, called the Somat machine, at the Hot Spot and the DūB. This increased the amount of food that could be composted at NAU. Since NAU had previously stopped using plastic utensils, styrofoam, and non-compostable items, this final step in composting eliminated the need for trash cans at these dining hall locations (Northern Arizona University, n.d.).

In 2019, a group of students focused their capstone project on investigating composting systems at NAU and determining whether in-house testing could replace external lab testing. Their study included a comprehensive management plan for Northern Arizona University's composting system and compared the costs of in house vs. external lab processing. Tests were performed on compost to determine concentrations of heavy metals, pH, temperature, ash content, E. coli, nitrate, ammonia, Salmonella, and the C:N ratio. Eight samples were taken from three piles at a height of four feet. The price of in-house testing was then compared to external labor costs. Their study concluded that using an external lab saves NAU ~\$64,000 over 20 years and ~\$5,400 per year in sampling costs (Almehmadi et al., 2019b).

In 2021, the Green Fund purchased and implemented a Subpod composting system at the SSLUG Garden on campus. The system used vermicomposting to increase compost output and student engagement. The intention was to conduct an experiment to see if composting rates could be sped up from 6-9 months using hot composting to 2-3 months using vermicomposting. It was scheduled to host a workshop every semester to expand the system, and a QR code was installed so students could scan it every time they dropped off food waste. This way, the number of people involved could be recorded. Currently, the POD is underutilized, and the reasons for this remain unclear.

Nevertheless, the Vermicompost Living Lab intends to reinvigorate this resource by hosting a workshop in partnership with SSLUG and independent research students focused on vermicomposting. The lab will contribute red wigglers that have been cultivated, and through the workshop, aims to enhance students' on campus awareness. The Green Fund also funded a composting bin at the Hotel & Restaurant Management building. The bin, located in the building's CULT Café, collects compostable waste from the Marriott's Demonstration Kitchen and the Cult Artisan Beverage Learning Café. The goal was to use this compost in the HRM kitchen garden that would be built in 2022 (Office of Sustainability 2022).

B. Current Composting Operations

NAU's two major dining facilities use Somat machines, which include a pulper, extractor, grinder, and dehydration unit. These systems process both pre- and post-consumer waste to produce a dehydrated compostable mixture. Food pre- and post-consumer is scraped into tubes or onto conveyor belts, which then transport the food into pipes and are carried through the Somat machine for processing. The organic waste is churned into an earthy byproduct. The final material is delivered to NAU Grounds at Howard's Mesa, where it is combined with carbon sources such as Ponderosa Pine chips, landscaping debris, and horse manure to create compost, used for campus landscaping, and shared with the Flagstaff community. In 2022, NAU composted 96,476 lbs of food alone. The University reportedly composts over 300,000 pounds of organic material each year, which prevents 312 metric tons of carbon dioxide from being emitted into the atmosphere. After installing composting machines, NAU saw a 90% reduction in food waste. To date, 1,500 pounds of post-consumer waste have accumulated per day at NAU (Green NAU, n.d. a).

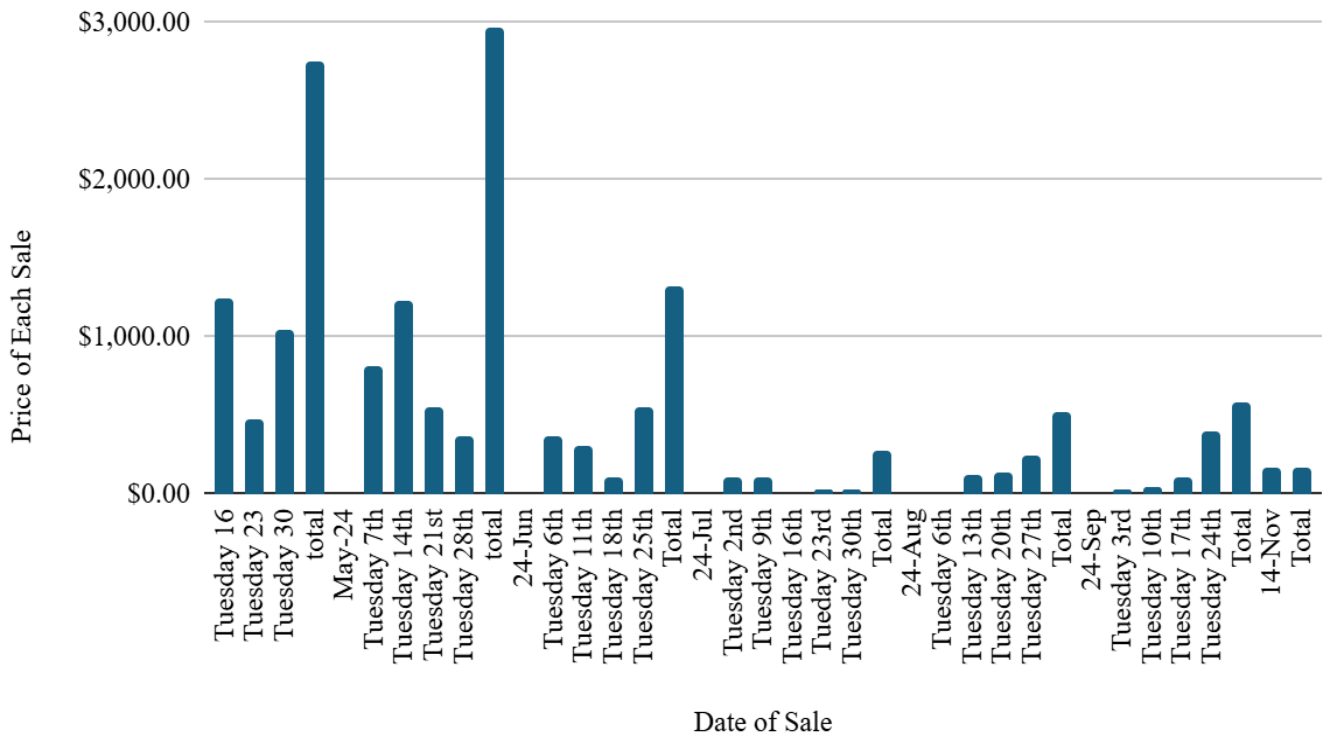


Transport technicians transport large bins of the Somat byproduct to the composting site located on Howard's Mesa. The composting site was named after Howard Coldwell for his passion and hard work in getting the composting system up and running. Coldwell noticed a shortage of quality soil on campus and began blending his own smaller mix that included the compost. He ended up with about 40 yards of blended soil. When his bosses noticed the product, they wanted to turn his passion project into a large-scale operation. This led to the incorporation of pre-consumer food from the dining hall into the composting system. After a year of successful composting, the Office of Sustainability Dining Hall Services purchased the Somat machine and began incorporating post-consumer food waste into the compost. Howard then uses tractors to mix the emulsified product with wood chips, cardboard, and dirt. Flagstaff soil has a high limestone content, making it hard for plants to grow. This nutrient dense composting soil supports a more bountiful harvest in agriculture.

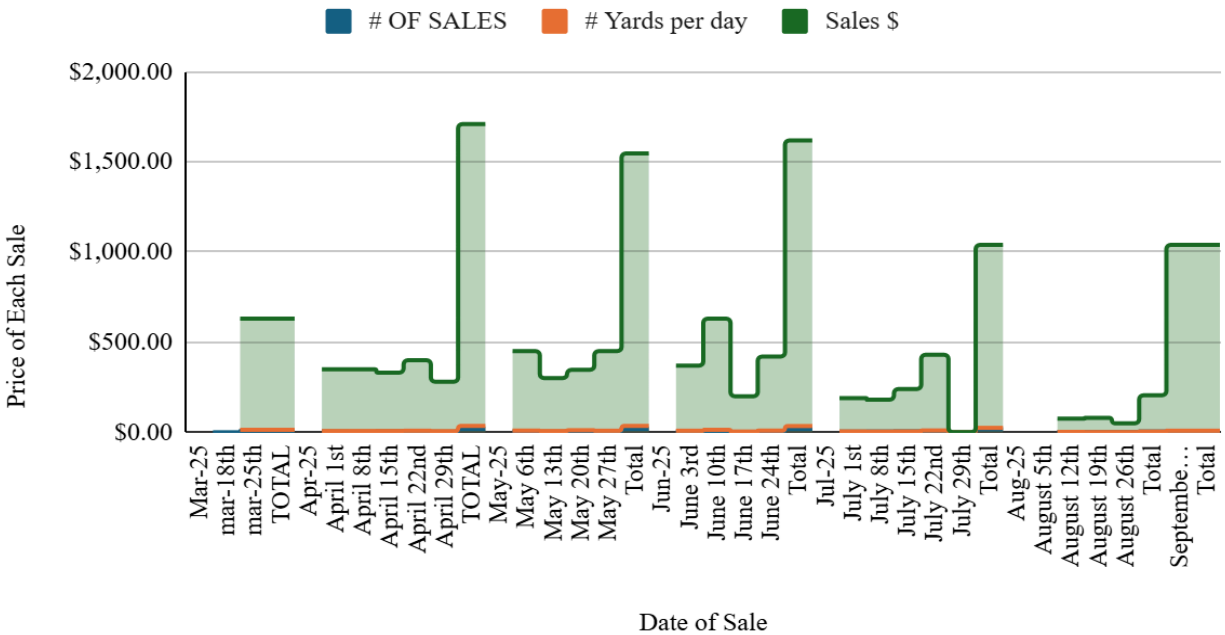
As of 2021, NAU made 4,500 yards of compost a year. Before this, Coldwell began giving the compost to community gardens, school groups, and in 2019 he was approved to start selling to outside customers. The result has been a program that has been functioning, but not reaching its full potential. In 2024, the total compost related sales at NAU was \$8,575. This number dropped slightly in 2025, with composting related sales totaling \$7,792. This represents a decline of \$783 dollars, or roughly 9.1%, which is relational to issues with operational challenges rather than a decrease in public interest. One of the key factors in the decline of composting sales is the inconsistency of the composting program. Since Howard Coldwell's retirement, a new composting manager has not been hired. According to Matthew Espino, the Landscape and Outdoor Services Manager on campus, the Grounds department is now responsible for caring for the compost. It is turned three times a week, adding in food from the dining halls. The problem is that the equipment operators who turn the compost have other duties

that they must attend to first, and the compost often gets left behind or turned inconsistently. The Grounds department and Property Surplus hold a materials sale every Tuesday. This includes compost, which is sold at \$50 dollars a yard. Espino also stated that in order to expand the current composting program, a budget position would need to be created for running the operation, plus an enclosed facility dedicated to storing compost.

GROUND S MATERIALS SALE FY'24/TOTAL SALES



GROUND MATERIALS SALE FY'25/TOTAL SALES



The sales across both years indicate irregular activity, including gaps in production and fluctuations in sales volume. Since this suggests that the system is not operating at full capacity, the sales likely underrepresent what the program could achieve if run efficiently on a full time scale. Even within these constraints, composting was still shown to be profitable on the market, with sales ranging from \$7,000 to almost \$9,000 dollars. By reinvigorating the current composting operations, plus adding in vermicompost, real potential lies in not only making a profit for NAU, but also offsetting the waste disposal/program costs. Traditional compost is often a less desirable product than vermicomposting. It is often sold in bulk at a lower cost, this means sales rely heavily on higher production. This is where vermicomposting plays a critical role in the expansion of environmentally friendly waste management on campus. Castings are a significantly higher selling soil amendment compared to traditional compost. They contain more readily available nutrients, support diverse microbial communities, improve soil structure and water retention, and are highly effective at improving overall soil health. This creates an opportunity for NAU to increase revenue without dramatically increasing production volume (Green NAU 2020).

C. Development of Vermicomposting at NAU: The Vermicompost Living Lab (VLL)

Although NAU's primary composting model has been mechanical and thermophilic, vermicomposting provides an alternative method of waste management that can support composting when the previous model is not working or to target smaller sources of organics, such as offices or residence halls. The VLL, funded by the NAU Green Fund in 2025, integrates waste management strategies to further reduce campus waste streams. The project provides not only research opportunities but also hands-on learning for students across different disciplines. Students have collected compost and worm-casting samples since its inception, which will be analyzed using various sequencing techniques (i.e., amplicon, metagenomics, metatranscriptomics) to identify key enzymes, genes, and taxa to better understand composting efficiency. Bioinformatics will also be used to analyze data using tools such as QIIME 2, which the Caporaso Lab in the Pathogen and Microbiome Institute develops and manages.

The lab itself sits at the Environmental Engineering Field Station on NAU's south campus, designed and developed under the guidance of Dr. Jeff Meilander. The project started in September of 2025, with an inground composting unit being built. The vermicompost houses Red Wiggler (*Eisenia fetida*) worms, which feed on organic food waste, with cardboard and paper as carbon sources to balance pH. Weekly data collection is conducted by the project's undergraduate research manager, including collecting compost and casting samples, and measuring temperature and pH. The worms are fed pre-consumer food collected weekly from the DüB Dining District on campus. This project serves as an investigative platform for sustainability practices at NAU and the need for expanded composting. Vermicomposting offers a cutting edge way to manage food waste sustainably. The product of vermicomposting is a nutrient dense soil that can then be used in campus gardens and landscaping or sold to external consumers at a profit.



The Vermicompost Living Lab also serves as a basis for community gatherings and outreach, hosting collaborative events with NAU's SSLUG Garden for educational and workshop purposes. The workshop will teach students how to make a vermicompost bin and how the nutrient cycle works in vermicomposting. The vermicompost bin created will then be donated to the SSLUG Garden to use. This will serve as a baseline for students to build their own vermicompost bins, then expand the capabilities of the SSLUG Garden. Outreach events also include participation in Earth Jam. The Vermicompost Living Lab will have an interactive info table where students can ask questions and get involved with the lab. Earth Jam is an annual Earth Day celebration that promotes sustainable practices by showcasing vendors, cultural performances, interactive exhibits, and live performances.

The development of this project has led to the creation of adjacent student-led research projects, such as that of undergraduate researchers Brooke Lercher and Andrew Grusenmeyer's project, "Worm Species Shape Gut and Vermicompost Microbiomes." This student-led project builds on the established Vermicompost Living Lab and investigates two worm species – *Eisenia fetida* (red wigglers) and *Lumbricus terrestris* (earthworms) – by comparing the relationships between their microbial communities and their efficiency in vermicompost systems. This work provides new insights into worm gut microbiomes, an area that has received limited prior research. Moreover, the operation of small-scale vermicompost systems helps promote campus-wide sustainability by diverting food waste, reducing emissions, and advancing NAU's Climate Action Plan.

III. Background Research on Vermicomposting Systems

A. Vermicomposting Fundamentals

Vermicomposting is a biological process that uses earthworms and microbial organisms to break down organic food waste under controlled temperature, moisture, and aeration conditions into a nutrient dense product made up of worm castings. Unlike conventional composting, which relies on thermophilic processes, vermicomposting results from the microbial interactions in the worm gut microbiome. Food waste, cardboard, and paper products are added to an aerated bin, where decomposition occurs, and the microbial process begins. Bacteria and fungi colonize waste products, partially softening and breaking down the organic matter. Worms are then drawn to this process as a food source and begin feeding on the decomposing waste. Worms are unable to easily digest intact polymers, making partially digested food from microbes more attractive. Upon intake of food, grinding of the material occurs in the gizzard, where enzymes break down macromolecules, and the microorganisms within the digestive tract further break down the organic material (Domínguez et al., 2019). During this process, worms fragment organic matter, increasing surface area and accelerating microbial decomposition rates. The gut of worms serves as a host to fungi, bacteria, and protozoa, which help digest complex molecules and continue decomposition after excretion (Kundariya et al., 2021). This process enhances decomposition and the transformation of nutrients into usable forms.

The microbial communities in vermicomposting are 10-20 times higher than those in common soil (Rinaldi, n.d.). Within vermicomposting, plant fibers are broken down via cellulose degradation. Within the worm gut, conditions are moist, buffered in pH, and highly enzymatic. After digestion, worms excrete waste called castings. These castings are high in microbial biomass and microbially enriched. The vermicompost product is substantially higher in enzymatic activity than the original substrate or common soil. Since microbial communities within castings continue to decompose after excretion, this enhances fertility when soil is mixed with vermicompost. One way this is done is through nitrification, the conversion of ammonium to nitrate, a form readily taken up by plants (Pathma & Sakthivel, 2012). Another important process in vermicomposting is humification, the formation of stable organic polymers from dead organic matter that resist further decay (Merriam-Webster n.d.). Through microbial decomposition, worms produce organic compounds that become more stable as humic substances. These substances, such as humic acids and fulvic acids, are carbon rich and have been shown to improve soil structure, increase water retention, and enhance nutrient holding capacity (Bogology, n.d.; Pathma & Sakthivel, 2012).



Image courtesy of Vuković et al. 2021.

It is important to aerate vermicompost and provide adequate moisture. This reduces the likelihood of pests and odors infiltrating the composting site. Fruit flies, fungus gnats, anaerobic conditions, and rodent infestations can result from overfeeding the worms. Providing too much food to the worms can overwhelm them. In order to maintain a neutral pH, adding a carbon source such as cardboard, egg cartons, or shredded paper is necessary every time the worms are fed.

B. Benefits to NAU

Incorporating vermicomposting as NAU contributes to Northern Arizona University's Climate Action Plan through providing an environmentally sustainable option to waste reduction and management. It directly supports the University's goal of carbon neutrality by reducing greenhouse gas emissions. One of the key goals of the plan is to reduce Scope 3 emissions, comprising emissions associated with material and food consumption and greenhouse gas emissions from waste management (Green NAU, n.d.a). Vermicompost directly addresses Scope 3 by redirecting pre- and post-consumer food waste away from landfills. In landfills, food waste

rots anaerobically, producing large amounts of methane, a key contributor to greenhouse gases. By comparison, vermicomposting is an aerobic process that reduces methane emissions by ~16-38% depending on the moisture conditions, ultimately contributing to NAU's carbon neutrality goal (Yasmin, 2022).

Northern Arizona University emphasizes maximizing resource conservation and efficiency as one of the steps towards carbon neutrality. Vermicomposting contributes to this goal by taking an continually reproducing byproduct (student food waste) and turning it into a usable substance. Vermicompost is a valuable resource for nutrient-dense soil that can be used in campus landscaping and gardens, or sold outside the University. The SSLUG garden on campus directly benefits from receiving this high quality soil amendment and has another resource for their garden composting program. Creating our own vermicompost on campus also reduces the need for outside fertilizers that can contain harmful chemicals, lowering the environmental impact associated with them. The Climate Action Plan also highlights the need to create a culture of sustainability.

NAU currently diverts pre- and post-consumer food waste from dining halls to on-campus compost piles at the South Campus compost facility. However, the system faces operational limitations due to inconsistent pile management, lack of dedicated staffing, and inputs containing meat, oils, and other materials that require sustained thermophilic conditions ($\geq 131^{\circ}\text{F}$) to ensure safe aerobic composting—conditions that depend on regular pile turning by trained personnel. At present, NAU Facilities does not have a dedicated compost manager, which results in food waste accumulating at the Howard Mesa site until staff time becomes available. These inputs also pose challenges for vermicomposting systems, as worms do not feed on them. As a result, decomposition is often uneven, piles attract pests (e.g., coyotes and ravens), and nutrient recovery is underutilized.

To strengthen NAU's commitment to sustainability and improve the campus waste stream, establishing a pre-consumer food waste diversion stream (fruit and vegetable scraps) specifically for vermicomposting offers an alternative pathway that is both practical and sustainable. This material is a high-quality feedstock for worms, and the system could operate on a predictable schedule coordinated between Facilities (Matt Espino), Sodexo sustainability manager (Jacob Vanetsky), and the vermicompost manager (Charlotte Hurley). Such a diversion pathway would alleviate some of the challenges in the current composting operation until additional funding becomes available to support a full-time compost manager.

Vermicomposting—the biological breakdown of organic matter by earthworms—also provides a fossil-fuel-free, low-cost, biologically driven strategy for reducing organic waste and producing soil amendments. This can create a small-footprint complement to the thermophilic composting operation at Howard's Mesa. Beyond processing pre-consumer waste, vermicomposting can also be used to further finish the thermophilic compost, helping ensure

pathogen reduction (particularly when meat-containing waste is inconsistently composted) and enhancing the final product with nutrient-rich worm castings. Worm biomass produced in this system could also be harvested, sold, or redistributed for additional vermicompost systems across campus to further amplify NAU's commitment to sustainability.

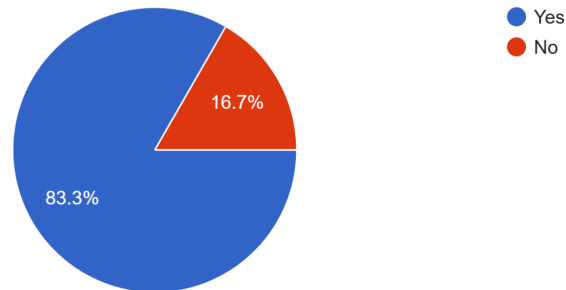
The implementation of vermicomposting systems at Northern Arizona University offers multiple quantifiable opportunities to reduce costs and recover resources across campus waste management operations. One of the most immediate savings comes from avoiding landfill tipping fees, which typically range from \$40–\$70 per ton of organic waste, depending on hauling contracts and regional landfill rates in Arizona (Gila County, n.d.). By diverting even a fraction of campus food waste away from landfill disposal, NAU can significantly reduce annual operational expenditures associated with waste export.

Additional savings are associated with reduced transportation and logistics costs. Organic waste processed on-site no longer requires frequent hauling to external composting or landfill facilities, thereby decreasing fuel consumption, vehicle maintenance, and driver labor hours. On a campus scale, this also reduces internal transport demands associated with centralized compost systems, particularly those that require repeated movement of heavy organic loads. Vermicomposting further reduces reliance on energy- and labor-intensive compost management equipment. Unlike thermophilic systems, which require regular turning using front-end loaders or windrow turners, vermicomposting systems operate with minimal mechanical input. This can translate into measurable reductions in equipment usage hours, maintenance costs, and associated fuel consumption.

There is also potential for generating revenue and offsetting costs. Vermicast is a high-value soil amendment that commonly retails between \$0.50–\$2.00 per pound, depending on processing and packaging, and can be used directly to reduce campus expenditures on commercial fertilizers for landscaping, green infrastructure, and research plots (Work Hard Worms, n.d.) Castings are significantly more expensive than standard compost, often ranging 5-10 times the cost per unit volume. Public interest surveys on the sale of castings indicate significant public interest. Over 83.3% of responses said that they have used compost, vermicast, or castings in their gardens before. When asked about the likelihood of buying castings, 50% said very likely, with 33.3% being unsure before receiving further information. Furthermore, 50% of responses requested additional information on purchasing castings. This data is promising and suggests avenues of revenue for the University.

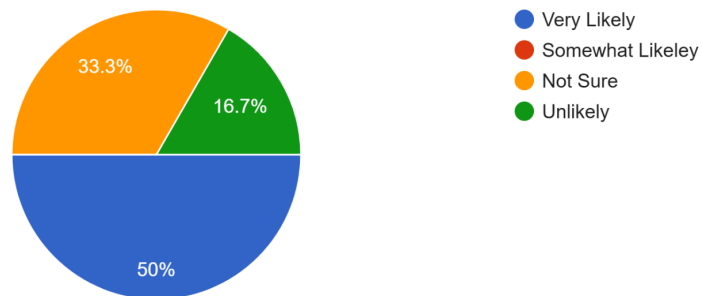
Have you ever used compost, vermicompost, or worm castings before in your garden or plants?
What was it like?

6 responses



How likely would you be to purchase worm castings if they were available locally?

6 responses



Finally, by substituting synthetic fertilizer inputs with vermicompost-derived amendments, the university can reduce recurring annual fertilizer procurement costs for grounds maintenance, while simultaneously improving soil structure and water retention in arid environments like Flagstaff. Collectively, these economic mechanisms position vermicomposting as both a cost-reduction strategy and a potential micro-revenue-generating system within a broader circular campus waste management model.

Vermicomposting benefits NAU in many ways. It directly advances the University's Climate Action Plan through waste diversion, resource efficiency, reduced greenhouse gas emissions, sustainability education, and community engagement. It provides innovative ways to reduce food waste on campus while fostering hands-on learning environments for students. This project bridges the gap between consumer waste and sustainability on campus.

C. Limitations of On-Campus Vermicomposting

Climate

Vermicomposting systems rely on mesophilic organisms, particularly *Eisenia fetida* (commonly known as red wigglers), which exhibit optimal metabolic activity within a defined temperature range. Temperatures outside the range of 3–27°C (55–80°F) result in decreased biological activity, and prolonged exposure to thermal extremes can lead to mortality. This presents a significant constraint in Flagstaff, where climatic conditions are highly variable due to elevation. Average winter temperatures frequently range between –9 and –6°C (15–20°F), while summer temperatures may reach approximately 28°C (82°F).

These seasonal fluctuations result in a marked reduction in earthworm activity during winter months, often for 4 to 5 months, thereby limiting the consistency of organic waste processing. Given the sensitivity of *E. fetida* to low temperatures, implementation of effective thermal management strategies is essential to maintain system functionality and organism survival during winter conditions.

Climate Adaptation Strategies

Seasonal variability can be mitigated through winterization of vermicomposting systems. This involves the application of insulating materials and structural modifications to reduce heat loss and retain metabolic heat generated by microbial and earthworm activity.

In this project, winterization was implemented by constructing a wooden enclosure around the vermicomposting pit, followed by the addition of layered insulating materials including tarps and straw. Structural support was provided using wooden beams (2×4 lumber), and an external tarp weighted with cinder blocks served as a protective barrier against snow accumulation and convective heat loss. Prior to system closure for winter, substantial quantities of organic material were incorporated to sustain biological activity. Data collection was temporarily suspended during this period. Additional strategies to enhance thermal stability include increasing the proportion of bedding materials (e.g., cardboard and paper), which improve insulation and contribute to maintaining a stable microenvironment within the system.

Feedstock Composition and Contamination

Effective vermicomposting requires careful control of feedstock composition. Suitable inputs primarily include plant-based materials, while the inclusion of meat, dairy, oils, and excessive citrus should be avoided. These materials can promote the proliferation of pests (e.g., rodents and maggots) and introduce pathogenic microorganisms.

Unlike thermophilic composting systems, vermicomposting does not achieve temperatures sufficient to ensure pathogen reduction through heat inactivation. Therefore, contamination with unsuitable organic materials or non-biodegradable components (e.g., plastics, chemical residues) poses a risk to both system integrity and worm health.

The initial implementation of the Vermicompost Living Lab used post-consumer food waste sourced from campus dining facilities. However, this material—processed via the Somat system—contained inputs including meat and dairy products. This resulted in infestation by maggots and the generation of odors associated with anaerobic decomposition, which subsequently attracted rodents. These conditions proved detrimental to earthworm viability.

This challenge highlighted the limitations of using post-consumer waste streams in vermicomposting systems. Although such materials can be effectively processed in thermophilic composting environments exceeding 55°C, they are unsuitable under mesophilic conditions.

Feedstock Management Strategies

To address these issues, the system transitioned to pre-consumer food waste collection. Collaboration with campus dining services, specifically through coordination with kitchen staff, enabled the consistent collection of uncontaminated organic material such as vegetable scraps and eggshells.

This process involved establishing a scheduled collection system aligned with food preparation activities. Weekly collection yielded approximately one six-inch-deep pan (≈ 2 ft in length) of suitable feedstock. Notably, dining staff indicated they could significantly increase this volume, suggesting scalability potential.

This shift not only improved vermicomposting system performance but also reduced reliance on energy-intensive processes, including mechanical waste processing, transportation, and thermophilic compost management. Importantly, post-consumer waste streams can continue to be diverted to existing thermophilic systems, allowing for an integrated waste management approach.

Labor and Spatial Considerations

Although vermicomposting is biologically efficient, processing rates are inherently slower than those observed in thermophilic systems. Earthworms typically consume organic material at a rate approximating 50% of their body mass per day, resulting in throughput limitations dependent on worm population size, environmental conditions, and system design.

Consequently, scaling vermicomposting operations to accommodate consistent waste streams would require either more distributed systems or larger, centralized units.

Vermicomposting systems also require more frequent monitoring and maintenance relative to mechanized composting systems. Key management activities include feedstock application, moisture regulation, temperature monitoring, data collection, and vermicast harvesting. While these tasks increase labor demands, overall mechanical input and associated energy use remain lower than in large-scale thermophilic systems.

Additionally, vermicomposting systems are better suited to continuous, moderate input streams rather than episodic influxes of large quantities of organic waste. Given the climatic constraints discussed previously, outdoor systems are subject to seasonal shutdown, necessitating the incorporation of indoor or protected flow-through systems. However, spatial limitations on campus may present challenges for implementation.

Operational and Programmatic Solutions

Expansion of vermicomposting infrastructure presents opportunities for both operational optimization and student engagement. Increasing the number of systems distributed across campus could mitigate climate-related constraints while reducing dependence on centralized infrastructure. Enclosed or semi-enclosed units, such as insulated sheds equipped with passive or solar-assisted heating, could maintain temperatures above freezing and support year-round biological activity.

From a labor perspective, expanded systems could support student employment and experiential learning opportunities. Funding mechanisms may include the sale of vermicast products, while student involvement could be facilitated through research projects, capstone experiences, volunteer programs, and interdisciplinary collaboration.

Temporal variability in vermicast production can be addressed by staggering system initiation and harvest cycles across multiple units. Additionally, pre-composting of feedstock prior to introduction into vermicomposting systems can enhance decomposition rates and improve substrate palatability for earthworms. While this approach requires additional space and infrastructure, it may significantly improve overall system efficiency.

E. Comparable University Programs

The Student Environmental Resource Center at the University of California, Berkeley has a student-led vermicomposting system that uses campus organic waste as compost for campus gardens. Founded in 2017, a vermicomposting team was formed to create a closed loop system that integrated campus pre-consumer waste to move towards a zero waste campus. Similar to the Vermicompost Living Lab at NAU, U.C. Berkeley coordinates with its Clark Kerr Dining Hall to collect pre-consumer food scraps, such as vegetable trimmings (Student Environmental Resource Center, n.d.). The project has reached a point where regular castings are harvested and used in campus gardens. Students can take a one unit course, Intro to Compost & Vermiculture, where

they learn about composting and how vermicomposting differs from traditional composting. Topics covered in the course curriculum include worm anatomy, composting science, the damage caused by artificial fertilizers, and how to build your own vermicompost bin at home. The faculty leading this course are members of the Student Environmental Resource Center, which supervises the ongoing management and harvesting of the vermicompost bins in the Clark Kerr Garden.

Texas State University is known for its applied learning curriculum, and vermicompost is a key component. A study was conducted at the University to investigate vermicomposting as a self-sustaining educational tool for environmental and waste management. The project included using Red Wiggler (*Eisenia fetida*) worms and cafeteria food waste to see if the system proved economical. Twenty five pounds of food waste were collected weekly from the university cafeterias, along with paper shreddings, as a carbon source. A small vermicomposting system was constructed inside a shed using recycled five gallon food service buckets. The worms were then checked two to three times weekly and eventually rotated throughout the system every four to five months. At the end of the study, vermicompost was packed into five gallon ziplock bags and either sold for profit or used in campus gardens. The economic analysis of the study showed the monetary value of using the castings as fertilizer and demonstrated that diverting the cost of waste disposal from production was cost effective (Montoya et al., 2016).

North Carolina State University is a leader in integrating vermicomposting systems and education. Extension specialist for solid waste management, Rhonda Sherman, runs the NC State University's vermicomposting program, offering a 4-H curriculum for children, industrial training, and resources for businesses. The 4-H curriculum includes six lectures on vermicomposting and the ecology of worms, a CD for visual references, and opportunities at NC State for children to learn firsthand about composting systems (Driscoll, 2024). In Raleigh, North Carolina, sits the NC State Compost Learning Lab (CL2), which serves as an educational, research, and demonstration site on the 1,500 acre Lake Wheeler Road Field Laboratory. The site features 25 types of backyard vermicomposting and composting units, ranging from small to medium scales. The site also has a 40 by 30 foot worm barn, which features a dozen different types of worm bins. These bins range from small household-sized vermicomposting bins to a 32 square foot continuous-flow-through raised vermicomposting bed (Sherman, 2025). On the roof of the worm barn is a rain garden that collects rainwater, which can then be used for gardens on campus. NC State periodically holds courses and educational events at the site for extension educators, farmers, recycling coordinators, entrepreneurs, food service members, gardeners, teachers, and the general public, featuring demonstrations on vermicomposting. NC State hosts an annual Vermiculture Conference, an event where attendees from around the world come to train on large-scale vermicomposting. Topics included developing proper feedstock, starting and maintaining composting, odor control, site selection, and marketing.

These Universities highlight a few ways that vermicomposting can benefit a college campus. The possibilities extend beyond composting itself to include higher education, economic value, community engagement, research, and sustainable practices. NAU would benefit greatly

from incorporating vermicompost as a permanent resident on its campus, with the potential to generate outside revenue and create opportunities for students. Vermicomposting would reduce campus waste while also opening the door to developing new curricula for classes, educational workshops, improved sustainability practices, expanded living lab operations, and economic opportunities. These models provide a template for NAU's potential expansion.

IV. Defining the Scale for Campus Operations

A. Scale of Systems (what would it look like as small, medium, large)

Scaling a vermicomposting system on NAU's campus requires analyzing the volume of available food waste, labor, and infrastructure capacity, as well as the program's current goals. Vermicomposting on campus has the potential to run large scale operations with the groundwork laid in small scale capacities. A small scale system could be run as a demonstration or pilot. Small bins could be placed around campus dorms or faculty spaces to test engagement with vermicomposting or serve as educational models in specific academic buildings. System design would include stackable trays or small tote bins. Locations would be offices, dorm buildings, classrooms, or in the SSLUG Garden greenhouse. The feedstock would be pre-consumer food, but would be brought in by faculty and students. Each bin could hold between ~1 and 10 pounds of worms. Since worms can process roughly half their body weight in food per day, this would result in roughly ~35 pounds of food processed per week on the higher end (10 lbs x 0.5 x 7 days).

Medium scale vermicomposting would result in a program sized operation or department led system. The Vermicompost Living Lab (V.L.L) provides a medium scale example, with the capacity to run a research project, and is managed primarily by an undergraduate research manager with the help of student interns. This scale serves as a functional campus program, providing a hands-on learning experience and offering a research opportunity into current composting conditions. The model for this would be in ground vermicompost pits or larger continuous flow through bins. You could place this model outdoors, at a dedicated site, or in campus gardens. This model can handle pre-consumer dining hall waste. Currently, the V.L.L is processing approximately 15 pounds, per week, at a successful rate. This scale is more waste reduction driven and can produce a product significant enough to be used on campus.

Large scale vermicomposting would significantly impact NAU's waste management and sustainability practices as they are designed for high-quantity waste diversion and can be integrated into a zero-waste strategy, including NAU's carbon neutrality goal. This format could be in large in ground vermicomposting pits or constructing a worm barn to house above ground flow through systems. To speed up the process of a system this size, existing thermophilic composting could be incorporated into the vermicomposting operation. Light pre-composting of food waste would accelerate decomposition for worms, enabling higher turnover rates. This scale would require dedicated site(s) to house the vermicomposting program, as well as additional workers. Long term, this scale provides the most benefit to NAU.

B. Evaluation Criteria

The feasibility and scalability of vermicomposting systems at Northern Arizona University can be assessed through a multi-criteria framework that integrates operational, biological, educational, environmental, and economic considerations. These criteria are intended to evaluate both system performance and broader institutional impact, while identifying potential constraints and opportunities for optimization.

Operational and Logistical Criteria

- **Feedstock capacity:** Assessment of system throughput relative to incoming organic waste streams, including the occurrence of processing bottlenecks, material accumulation, or disruptions in collection and delivery.
- **Labor requirements:** Evaluation of staffing needs, including the feasibility of relying on student or volunteer labor versus the necessity for dedicated personnel to ensure consistent system operation.
- **Odor and pest management:** Monitoring of odor production and pest presence, alongside the effectiveness of mitigation strategies across different system scales.

Biological Performance Criteria

- **Earthworm population dynamics:** Measurement of survival rates, reproductive success, and biomass accumulation over time to determine system stability and potential for population expansion.
- **Vermicast quality:** Analysis of physicochemical and biological properties of produced castings, including nutrient content (N, P, K), microbial diversity and activity, pH, and moisture content.
- **Processing efficiency:** Quantification of the time required to convert feedstock into stabilized vermicast, requiring consistent data collection on input and output rates.

Educational and Research Value

- **Undergraduate research integration:** Evaluation of student participation, including the number of students engaged in research, independent study, or applied learning through the system.
- **Data collection infrastructure:** Feasibility of implementing and maintaining monitoring systems (e.g., temperature, moisture, pH, CO₂ flux, and microbial sampling), particularly in support of broader climate and sustainability initiatives.
- **Demonstration and outreach value:** Accessibility and visibility of vermicomposting systems for educational tours, community engagement, and outreach activities, including tracking of volunteer participation and engagement hours.

Sustainability and Campus Impact

- **Waste diversion rate:** Quantification of organic waste diverted from landfill on an annual basis.
- **Compost production and utilization:** Measurement of vermicast output and evaluation of the campus's capacity to utilize this material in landscaping, research, or food production systems.
- **Carbon footprint:** Comparative assessment of greenhouse gas emissions avoided through waste diversion relative to emissions and energy inputs associated with system operation.
- **Alignment with institutional sustainability goals:** Degree of integration with existing waste management infrastructure and consistency with campus climate action and sustainability commitments.

Economic Criteria

- **Operational costs:** Ongoing expenditures associated with labor, materials, and system maintenance, and their long-term financial sustainability.
- **Revenue and cost-offset potential:** Evaluation of potential income streams, including the sale of vermicast, vermicompost tea, or earthworm biomass, as well as opportunities for external funding and grant support.
- **Cost efficiency:** Calculation of cost per unit of waste diverted (e.g., cost per ton), enabling comparison across system scales and with alternative waste management strategies.

1. Scale of Production Breakdown

Worm's process roughly ~25-50 percent of their body weight depending on the season. With 1,000 worms weighing a total of 1 pound as the standard starting amount for small vermicomposting bins, this means one can expect roughly 2-3.5 pounds of food per week to be processed (Environmental Protection Agency, n.d.). With a project this small, only one to three workers, in the form of student interns or researchers, would be needed per week to maintain the bins, depending on the number of bins. Typical vermicomposting bins are around 20 to 40 gallons or 2 to 4 feet in length, requiring a small amount of indoor space. With the median output of these bins being 3 pounds per week, the scale is relatively small compared to the amount of food waste produced on campus. These bins would better serve as personal bins for dorms and faculty members rather than the dining halls. An alternative to expansion would be to install small vermicompost bins around campus, with a main outdoor vermicomposting site. This could serve as an overflow site or as the origin for small bin materials. Commercial small bins range from \$80 to \$120, depending on shipping (Matney, 2022). Inexpensive DIY options are available, with 27 gallon storage totes starting at \$10 at a retail hardware store. Modifications will be needed on the DIY route, requiring more time and energy to complete the project. The

annual costs of small scale vermicomposting are mostly tied to labor, including compensation for students or faculty for maintenance, with occasional repairs potentially needed.

To increase output would require upscaling the system. A medium sized vermicomposting pit could be dedicated to composting pre-consumer food from one dining hall on campus. For example, if a medium-sized system had ~50-100 pounds of worms, then at max output, production could reach ~350 pounds of food processed per week ($100 \times 0.5 \times 7$). The number of workers required would also increase, likely needing 3-6 student interns to maintain the vermicomposting system, collect data, and coordinate foodstock pickup. This operation would also require a faculty advisor to oversee production and coordinate with the university for site location. This system could be either indoors or outdoors. Due to the capacity of medium scale systems, it is suggested that this be done in a continuous-flow-through bin if frequent collection is desired. It can be done in the ground, but it may require more coordination for bi-yearly product collection, which could disturb the worms' environment. Depending on the system selected, this calculates to roughly 10-20 pounds of waste per square foot per week, resulting in approximately 17.5 square feet needed for the ~350 pound example. Costs for this scale vary significantly, with medium sized store bought flow through bins around ~\$1,750-\$10,000 to in ground systems requiring mostly labor costs at \$15.75 an hour for students, along with structural costs (Hiwassee Products, 2025). Money can be made back starting with this scale by selling castings at \$2 a pound (Financial Models Lab, 2026).

Large scale systems are desirable for institutional wide waste management. This scale could process ~1,000-20,000 pounds of waste per week, requiring thousands of pounds of worms. Large scale systems are usually operated in either continuous flow through bins or in long windrows. A major difference between these operations and smaller scale systems is that pre-composting is a key element. Three to six weeks of pre-composting are used to stabilize food systems and avoid shocking the worms (Herlihy, n.d.). To run things efficiently and properly, a full time manager would be needed, with 10+ part time students or staff. This scale should be treated as a professional operation on campus, with proper training and safety measures in place. Site size would greatly increase, and the need for its own outdoor space with proper winter coverings would be taken into consideration. This is arguably the most efficient system, with long term impacts, but requires the most mechanization and management of the three scales.

NAU would benefit greatly from making large-scale vermicomposting an integral part of its waste management system, given that over 300,000 pounds of food waste are produced annually on campus. This system could work in tandem with improvements to the preexisting composting mechanisms, creating a second area for pre-consumer waste. This would be the most costly size, but with the greatest output. In India, a study was carried out to analyze the cost benefit of large scale farm vermicomposting on a college campus. The study, conducted in 2021 at the Professor Jayashankar Telangana Agricultural University in Hyderabad, concluded that the

total cost of producing vermicompost per annum was 7,25,805 Rupees. When converted to U.S dollars, this resulted in a total cost of \$7,652.12. That being said, the reported study reported “The total revenue earned per year was Rs. 10,22,400 and Rs. 16,74,000 at PJTSAU and market prices, respectively. Net returns and B: C ratio values indicated that the vermicompost production unit was economically viable at both the prices” (Satish, 2023). In this study there was a profit of almost 3 million dollars. The paper went on to acknowledge the challenges of vermicomposting, including water scarcity in summer months, worm deaths due to adverse weather conditions, and labor costs. NAU does not need a production of this scale, but it does show that the project was well worth the cost. NAU could expand to a similar but less extensive large-scale production and use this example as a guideline.

2. Long-Term Sustainability

For long term sustainability of this project to be achieved at NAU, vermicomposting must become an integral part of campus waste management. When driven by data-backed research, reliable systems, and proper institutional management, vermicomposting has the potential to improve how campus food waste is processed. Operational reliability includes Flagstaff’s climate and the appropriate measures in place to account for extreme winters. Winter temperatures in northern Arizona regularly drop below freezing, which can significantly decrease production or be lethal to worms. Since vermicomposting is a mesophilic process, it requires moderate to warm temperatures to function properly. This can be compensated for by housing the vermicomposting structures indoors, in greenhouses, or outside with insulation built around the vermicompost. This factor increases seasonal unreliability unless internal temperatures are controlled and monitored. Another method is to reduce feeding rates in winter to avoid overstimulating the worms, while using summer as peak production months. This method is less desirable on campus because the busiest period for food waste is September through May. Adaptation to adverse weather aligns with the NAU's Climate Action Plan, which emphasizes resilience and adaptability to environmental conditions (Green NAU, n.d.b). A controlled vermicomposting project directly supports this goal by increasing resilience in campus operations instead of outsourcing to external waste systems.

Maintaining a labor force for campus vermicomposting is challenging, as student turnover occurs every 1-4 years. This can lead to a loss of institutional knowledge about the systems in place if communication between hirings is not sustained. It would be wise to keep a consistent vermicomposting general manager as a full time employee, serving as the head of operations and familiar with running things. Repeated and consistent training provides a structured student employee program that remains repeatable despite higher turnover rates. A course manual for student researchers could be created, and a system of operations guidelines for each new employee could be followed. To expand knowledge beyond those directly employed by the program, vermicomposting could be woven into course curricula. Field trips to the site could be coordinated, or a sustainability unit could be dedicated to learning about vermicomposting at

NAU. Internships and student research assistant roles could be created to spread the workload out. The Vermicompost Living Lab on campus provides hands-on experiences for students who want to either be employed by the University or earn class credit through independent study, allowing them to serve as operators and researchers.

Long-term sustainability relies on steady streams of income rather than one time grants. Costs include startup materials such as worms (initial and replacement), physical infrastructure (bins, outdoor greenhouses, insulation materials), laborers (student workers and faculty), and maintenance (tools, bedding materials (cardboard or paper), repairs). One in house source of revenue is the Green Fund, which provides funding for sustainability projects through student fees. The Green Fund is responsible for funding the Vermicompost Living Lab. A grant proposal could be written for the expansion of the VLL, with upkeep costs covered by the sale of castings and finished vermicompost. A vermicomposting club could be founded with monthly member fees collected that go towards student projects. For student researchers and graduate students, research grants are available through the USDA, EPA, and specialized agricultural programs focusing on waste management, food waste reduction, sustainable soil management, and urban agriculture (Gedert, n.d.).

It is important that vermicomposting brings in data collection and research to support longevity through academic contributions. The system is made measurable by tracking how many pounds of foodstock are given and processed per week, worm population growth and decline through random population sampling, composting output of castings, labor hours recorded each week, and cost vs. savings analysis annually. Laborer hours should also be tracked to compare with system productivity and the opportunity for expansion. The NAU climate plan emphasizes tracking emissions and resource use to move towards carbon neutrality. As discussed in a previous section, vermicomposting research can help reduce Scope 3 emissions by reducing food waste. A collection of maintenance data (pH, temperature, moisture levels, worm castings, soil samples) can contribute to research datasets for student sustainability studies. Research can be conducted as a capstone project, graduate research, or an independent study.

Institutional backing is crucial for long term success. When run solely on student enthusiasm, projects fall apart once student leaders graduate. The program must be incorporated into current operations before employee turnover. Funding also needs to be institutionally backed, even if the University is not fiscally responsible, to avoid a lapse in funding. For example, the Green Fund is an institutionally run organization that students financially support. Commitment to long term sustainability of vermicomposting at NAU requires consistent management, integration into dining hall services, inclusion in sustainability practices, and consistent financial support.

3. Institutional Fit

NAU's Climate Action Plan focuses on reducing emissions, minimizing campus waste, and creating a culture of sustainability. Vermicompost contributes to this plan by diverting food waste from landfills, thereby reducing methane, a greenhouse gas. Vermicompost supports local community building and soil health by providing an alternative to conventional fertilizers, which can harm our waterways. The gardens at NAU can be used as a teaching ground for nutrient cycling and the importance of protecting our waterways, with vermicomposting being an alternative to products with harmful chemicals. In 1973, NAU became one of the first major universities to offer environmental science as a degree, and today, over two-thirds of the University's academic departments offer sustainability courses (Green NAU, n.d.c). Vermicomposting can be integrated into the environmental sciences curriculum through renewable methods for recycling waste while saving water. It can also be used to teach soil science or microbial ecology in the Biology Department. The Engineering Department could collaborate with the Science Department to design vermicomposting waste systems on campus. The Climate Action Plan has been largely facilitated by Abraham (Avi) Henn, NAU's manager of sustainability. Henn believes in seeing NAU as a "living laboratory", with the opportunity to learn things through trial and error, using intellect and academic resources to get important work done (NAU Stories, 2023). Henn's philosophy is in this project's name. The Vermicompost Living Lab promotes learning through first hand experimentation with natural resources. It takes what would otherwise be sent as waste to a landfill and instead creates a product that benefits multiple communities. It is an exemplary system that produces real data and real change.

Cross department collaboration is promoted by combining departments to achieve the desired end goal of vermicomposting becoming a key player in NAU's waste reduction strategies. The Office of Sustainability serves as a framework for sustainability goals within the Climate Action Plan. It incorporates previous composting practices to improve the overall system and make it more efficient. Collaboration with Dining Services will be needed to facilitate food waste collection for pre-consumer foodstock. Facilities or Grounds Crew can help with maintenance of the project site and collect the finished product for use as fertilizer. The Green Jacks are a student-run organization that promotes sustainability on campus. Collaborating with them could lead to sponsorship or serve as a volunteer hub for maintenance of the vermicompost. NAU already supports student sustainability groups, food recovery programs, and campus gardens through clubs such as the NAU Environmental Caucus, O2Go, and the Food Recovery Network (ASNAU, n.d.). The Vermicompost Living Lab and its future expansion would be a perfect addition for groups to work with. Collaboration matters because it supplies a consistent feedstock, shares labor and responsibility, and integrates education into the operation.

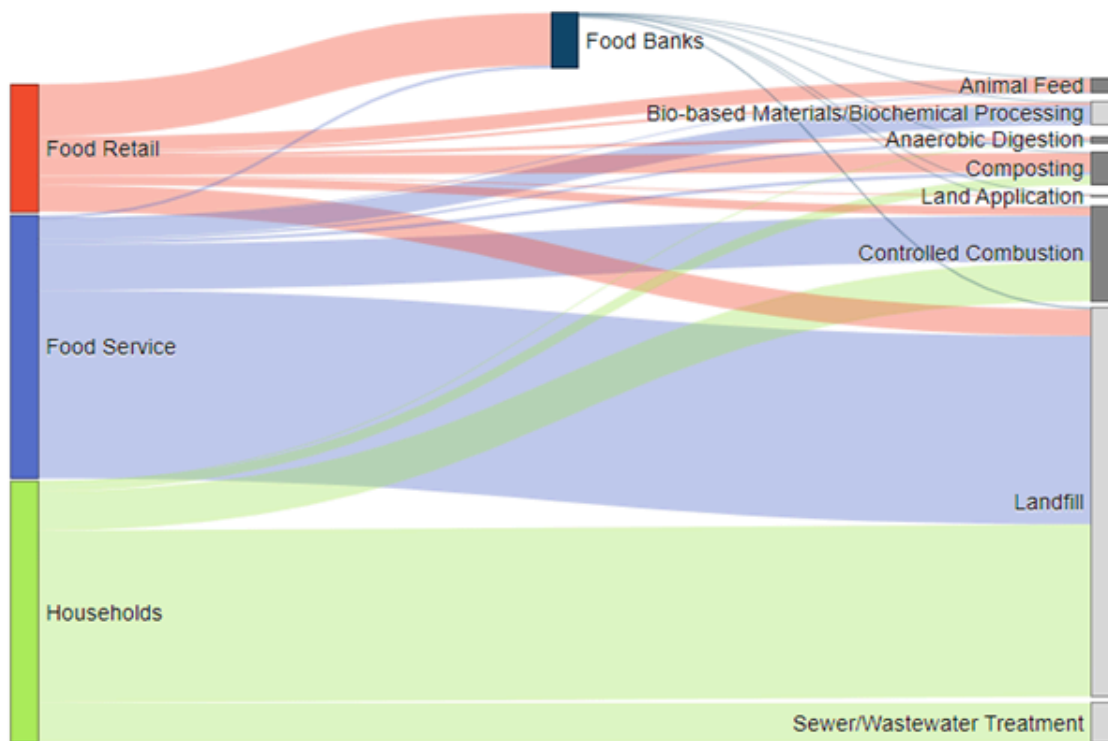
Vermicomposting at NAU can be a feasible long-term solution for reducing campus waste when the above factors are incorporated. When institutionally supported, the project can serve as a data-driven learning hub that benefits students and the community through hands-on

education. This project is a multipurpose tool that will reduce waste emissions, connect students across different disciplines, advance research, and strengthen core campus values.

V. Potential Environmental, Educational, and Research Benefits

A. Environmental Benefits- recap from long term sustainability, NAU goals, what vermicompost does

One of the most immediate benefits of vermicomposting is its ability to divert waste away from landfills such as Cinder Lake Landfill. Food waste is a major contributor to U.S. landfills, with the EPA estimating in 2019 that 66 million tons of food waste were generated from the food retail, food service, and residential sectors (Environmental Protection Agency, 2019). Of this waste, about 60% was sent to landfills. Another 40 million tons of food waste was generated in the food processing sectors. The following graph shows the flow of wasted food in food service, food retail, and residential sectors to the management pathways:



WASTED FOOD GENERATION AND MANAGEMENT FLOWS (2019)

The following table shows how food waste has been managed since 1960. A dash in the table means data is not available.

1960-2019 Data on Food in Municipal Solid Waste by Weight (in thousands of U.S. tons)

Management Pathway	1960	1970	1980	1990	2000	2005	2010	2015	2017	2018	2019
Generation	12,200	12,800	13,000	23,860	30,700	32,930	35,740	39,730	40,670	63,130	66,220
Recycling	0	0	0	0	0	0	0	0	0	0	0
Composted	-	-	-	-	680	690	970	2,100	2,570	2,590	3,300
Other Food Management	-	-	-	-	-	-	-	-	-	17,710	13,640
Combustion with Energy Recovery	-	50	260	4,060	5,820	5,870	6,150	7,380	7,470	7,550	9,650
Landfilled	12,200	12,750	12,740	19,800	24,200	26,370	28,620	30,250	30,630	35,280	39,620

*Note that 2018 was the first year for which EPA used a new methodology, so the generation estimates increased due to the expanded scope of the new methodology.

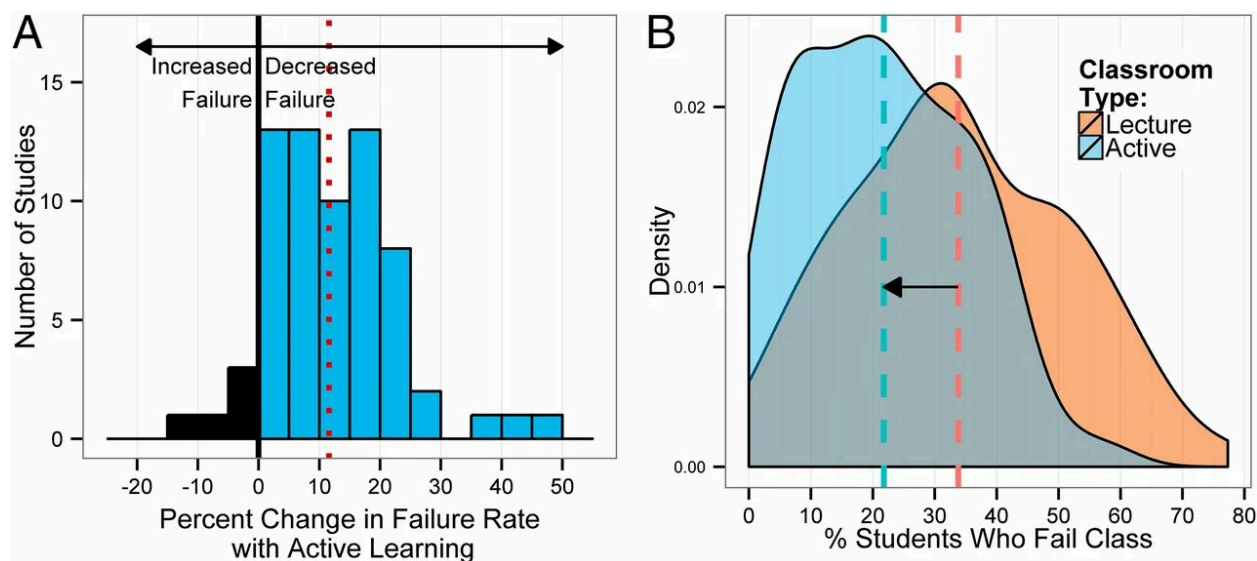
Image courtesy of the Environmental Protection Agency, 2019.

The data shows that the largest share of food waste has been sent to landfills since 1960. Food waste is the leading material in U.S landfills, making up approximately 24% of municipal waste (U.S. Food and Drug Administration [FDA], 2025). Due to quick anaerobic decomposition, landfilled food disproportionately generates methane, a potent greenhouse gas. Methane is 25 times more potent than carbon dioxide over 100 years. Landfilled food accounts for 58% of all landfill methane emissions (FDA, 2025). The U.S has established a goal to reduce food waste 50% by 2030 through the collaborative efforts of the EPA, FDA, and USDA. In Flagstaff, Cinder Lake Landfill handles approximately 385 tons of solid waste daily. Given that food waste accounts for nearly a quarter of total waste, a significant portion of this is likely food waste. At NAU, over 90% of food waste is currently composted, with over 300,000 pounds composted annually. The composting program was a resounding success under Howard Coldwell's leadership. Since his retirement, the program has suffered, and most post-consumer food waste is no longer properly composted. Vermicomposting on campus can expand NAU's composting capabilities by introducing new methods to revitalize the program.

Vermicomposting enhances soil diversity by producing microbial-rich, nutrient-dense castings, thereby improving soil structure and fertility. Vermicompost also improves soil water retention, reducing the need for excessive watering. At NAU castings could be used for campus gardens, landscaping, and research projects. This creates a closed-loop system that reduces the need for synthetic fertilizers. The Haber-Bosch process is used to create nitrogen fertilizers that support ~50% of the world's population through increasing food production, but it is highly energy intensive. This process accounts for ~3-5% of global natural gas consumption and ~1-3% of CO₂ emissions. Vermicompost provides a natural alternative by slowly releasing nutrients and supporting beneficial microbes without using excess energy.

B. Educational Benefits

Vermicomposting turns waste management into a hands-on learning tool, in line with NAU's "living lab" model. These hands-on experiences can be used by students from diverse backgrounds, including biology, environmental science, engineering, and hotel/restaurant management. Environmental science students will get experience with waste management and sustainability. Biology students will learn about decomposition and microbial ecology, while engineering students gain experience in system design and processing. Hotel/restaurant manager students can work directly with the dining halls to brainstorm food waste reduction strategies. NAU emphasizes experiential learning through the Office of Sustainability, which serves as a hub for sustainability operations. Applied learning has been shown to improve retention of scientific concepts and increase student engagement (Freeman et al., 2014). Studies have suggested that active learning increases examination performance by an average of half a letter grade and increases failure rates by 55% when taught in traditional lecture settings. To test the hypothesis that lecturing maximizes learning outcomes and student performance, researchers from the University of Washington conducted a meta-analysis of 225 studies reporting examination scores and failure rates among undergraduates in STEM (science, technology, engineering, and mathematics) degrees who learned in traditional lecture settings vs. active learning. The results indicated that examination scores improved by roughly 6% in active learning scenarios, and that students in traditional lecturing environments were 1.5 times more likely to fail. Heterogeneity analysis concluded that both statements held true across all STEM disciplines and that the greatest effects were observed in small class sizes. However, they still held true across all class sizes (Freeman et al., 2014). Below are two figures that display this data.



Changes in failure rate. (A) Data plotted as percent change in failure rate in the same course, under active learning versus lecturing. The mean change (12%) is indicated by the dashed vertical line. (B)

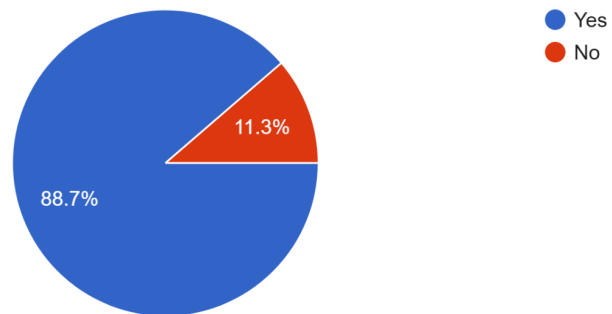
Kernel density plots of failure rates under active learning and under lecturing. The mean failure rates under each classroom type (21.8% and 33.8%) are shown by dashed vertical lines.

The Vermicompost Living Lab (V.L.L) and its expansion would provide the active learning opportunities needed for student success. The V.L.L provides ongoing opportunities for students to engage in capstone projects, outreach, workshops, and research.

Surveys conducted across campus departments indicated strong interest in vermicomposting among students and faculty. An email was sent to various departments and classrooms with a survey link to a vermicomposting questionnaire. A total of 276 responses were collected, with only 29.3% having definitely heard of vermicomposting before, while 88.7% want to learn more about it. After a brief introduction to vermicomposting, 84.6% of student answerers said yes to wanting a vermicomposting bin in their dorm. Among employees, 75% said they were interested in having a vermicomposting bin in their office space. On a campus-wide level, over 96% said they would participate in vermicomposting if it were implemented on campus. In the questions and concerns free-response section of the survey, the majority of responses indicated that they had no concerns or wanted to know more about how to get involved in facilitating this on campus. Some concerns included proper use or potential smell, which, with proper education and a dedicated management team, can be easily combated. These responses show that students and faculty alike strongly desire to expand waste-management services on campus. One student felt inspired to get involved or further vermicomposting outside of campus housing, quoted as saying, “I do not live on campus but in student housing near campus. I live in a studio apartment, but there is a community garden nearby that I would be happy to reach out to and see if there’s a way to implement vermicomposting. Additionally, I could contact my apartment complex about the possibility of starting a vermicomposting initiative. I’ve always found it a bit strange to throw food scraps into the trash. Growing up in a rural area, we would either feed scraps to our animals or incorporate them back into the soil for our crops.”

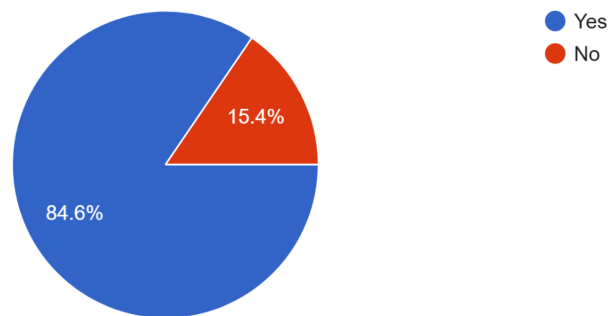
Would you be interested in learning more about vermicompost?

274 responses



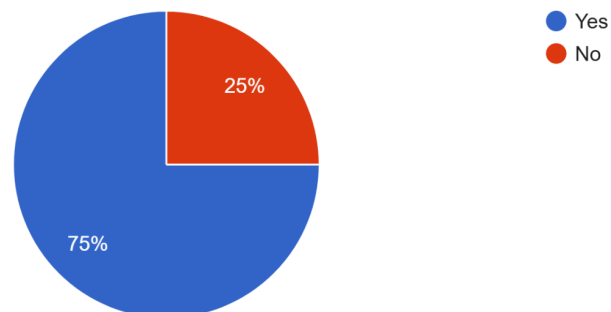
Would you be interested in having a vermicompost bin in your dorm space (if applicable)?

259 responses



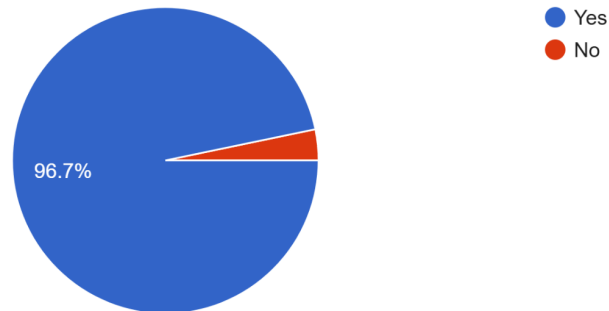
Would you be interested in having a vermicompost bin in your office space (if applicable)?

232 responses



If vermicomposting bins were implemented around campus, would you participate in putting your food scraps into them?

276 responses



Personal Testimonials

Charlotte Hurley-

So far this school year, the V.L.L has allowed student researchers, such as myself, to gain experience in the field, a coveted role before graduation. This job has changed the trajectory of my academic career. It has allowed me to gain hands-on research experience. It created opportunities to practice leadership by creating and leading workshops. It has also provided me the opportunity to create my own poster for the first time, which was presented in the Undergraduate Symposium at NAU. I was able to attend conferences, meet incredible people, and network with other student researchers. Before this project, I was very scared of graduating and not having enough on paper to show for myself or be hired. Workplaces want some form of prior experience, and when working full-time while being a full time student, it is hard to fit that in. This research project provided that experience while also teaching me a lot about myself.

It also sparked the creation of branching experiments, such as the Green Fund sponsored project run by Brooke Lercher and Andrew Grusenmeyer. Vermicomposting projects naturally support community engagement and a culture of sustainability through student workshops and community outreach. The V.L.L worked alongside Paige Yeakle from the NAU SSLUG (Students for Sustainable Living and Urban Gardening) Garden to create an outreach event and vermicomposting workshop. Paige and I are now friends, and this project brought that friendship together. Leading this workshop taught me about event organizing and engagement while also teaching participants about vermicompost and its importance. We then had participants engage in

hands-on work building two vermicomposting bins for the SSLUG Garden. The Vermicompost Living Lab could become a campus staple, serving as a central research and education site for students. The site could function as a demonstration space for students, encouraging lasting student involvement. Eventual expansion of the program could include high school internships and pre-college summer programs. This would build interest in sustainability careers and bolster students' college applications by sharing personal experiences with creating change.

Brooke Lercher -

The V.L.L enticed my interest in vermicomposting systems and prompted me to submit a grant proposal to the NAU Green Fund to support further vermicomposting research. Following the approval of this grant proposal, I became deeply involved in undergraduate research through employment at the Pathogen and Microbiome Institute in the Caporaso Lab. This hands-on experience allowed me to apply key concepts learned in previous science courses while collaborating with other student researchers, including Andrew Grusenmeyer. In the lab, I obtained valuable experience handling specimens, maintaining multiple vermicompost systems, and collecting and analyzing samples. I also developed a deeper understanding of sustainable waste management, particularly through the biological breakdown of food scraps into nutrient rich compost. Through this research, I explored the digestive process of worms, specifically investigating the relationship between microbial communities within worm guts and their efficiency in vermicompost systems, which strengthened my interest in environmental microbiology. In addition to technical skills, I also improved my scientific communication and presentation skills by sharing my research findings with the broader NAU community. Likewise, I participated in multiple student-outreach events to encourage campus involvement in diverting food waste and reducing emissions through the utilization of vermicompost systems. My involvement and interest, starting with the V.L.L., provided me with additional research opportunities during the summer and the following school year, allowing me to build on my previous work and continue promoting campus sustainability.

Aisle Herrera-

Being involved in the vermicomposting project has had a really positive impact on me. It gave me valuable experience and helped build my outreach skills. I reached out to faculty and departments, clearly communicating information. I've also learned what it's like to work as part of a team on a larger scale project. This has made me more confident and comfortable in contributing to research. This project has also strengthened my research interest. Specifically, seeing how an idea can be turned into a large-scale project. I also enjoyed knowing my work was

useful and that I was contributing to the team. Overall, I'm glad I was allowed to be a part of the vermicomposting project.

C. Research Benefits

Vermicomposting offers ample research opportunities for students to study soil biology, waste decomposition, microbial ecology, plant diseases, agriculture, organic chemistry, nutrient cycling, and related fields. It creates opportunities for graduate theses, student research, and faculty-led projects. It also incorporates engineering through bin design, climate adaptation, and long term campus sustainability datasets. A key factor in vermicompost research is soil microbiology and microbial ecology. Worms physically break down foodstock, but the microorganisms they work with are responsible for the biological conversion into a nutrient dense product. Previous research shows that vermicompost supports diverse microbial communities, promotes plant growth by promoting bacterial growth, and enhances nutrient cycling through enzymes (Pathma & Sakthivel, 2012). Some of the beneficial properties of vermicompost may be attributed to the changes in the composition and functions of bacterial communities during vermicomposting (Domínguez et al., 2019).

Despite increasing interest and research into vermicomposting over the past few years, knowledge gaps remain, and further research is needed. This provides an opportunity for NAU to be a forerunner in the biotechnological process of vermicompost. There have been challenges in clarifying and understanding the mechanisms involved in the vermicomposting process. Gaps include a lack of a deeper understanding of bacterial composition, the amount of vermicompost, the effects on heavy metal content, organic waste selection, plant pathogens, and diseases (Vuković et al., 2021). Additional knowledge of the role of bacterial communities is required. Although previous research exists, further research into bacterial succession and the dynamics between bacteria and specific earthworms could yield a desired final product with specific characteristics. Furthermore, knowledge of the microbial communities in both earthworms' guts and soil could lead to a better understanding of decomposition processes and phases. Studying the bacteria most active at different stages of vermicomposting would enable the most efficient use of the process and help predict the final product.

Another area for further research is determining the optimal amount of vermicompost to add for soil function. A student research experiment could be enacted to determine how long it takes for different amounts of vermicompost to lose its positive effects. The mechanisms by which high or low levels of vermicompost affect plant species can be further investigated. Most of the previously published literature focuses on the negative effects of hormone-like molecules, plant stress from high nutrient levels, nutrient competition with surrounding plants, and physical changes in soil (Blouin et al., 2019). This provides an opportunity for student-led research.

Vermicompost can potentially affect the bioavailability and mobility of heavy metals in soil. There has been a suggestion that vermicomposting could reduce heavy metal concentrations

in soil, but further research is needed to examine the connection between heavy metal fractions, earthworms, and heavy metal redistribution (Swati & Hait, 2017). That said, it is clear that positive effects are present, and that vermicomposting, when used under the right conditions, can contribute to heavy metal remediation. The lack of understanding here is how exactly that happens. Experiments and examinations into this topic open the door to direct or indirect effects of heavy metals in soil and, therefore, on human health. It could also lead to experiments into nutrient modification. For example, if some populations have higher rates of zinc deficiency linked directly to human health, an alternative to supplementation could be to increase soil zinc concentrations. Not only does this area provide a research opportunity for NAU, but it also contributes to sustainability and food goals. An on-campus food garden could be created that specifically engineers vermicompost with the nutrients that students who may otherwise not have access to need.

At NAU, a student-led research project is underway in which students are experimenting with different types of organic food waste used in vermicomposting. This project could expand its parameters beyond determining the most beneficial foodstock for vermicompost to include investigations of the final product. If the characteristics of the initial substrate are cataloged, components of the final product could be used to create seedlings for similar substrates. Similar to the heavy metal investigation, vermicompost could be engineered into different substrates to be most beneficial for different environments. If NAU can investigate these knowledge gaps, it would enable further utilization of the vermicomposting process.

VI. Preliminary Limitations, Risks, and Knowledge Gaps

A. Resource & Infrastructure Constraints

Vermicomposting at NAU requires navigating physical infrastructure limitations, including construction requirements at NAU and Flagstaff's climate. To ensure a system that is both scalable and long-lasting, certain precautions must be taken. One of the most critical factors in vermicomposting at NAU is considering Flagstaff's climate. This means there needs to be a temperature-control element in the vermicomposting system to achieve year-round production, or to pause during the winter months to account for the slower turnover rate. Vermicomposting relies on Red Wiggler worms (*Eisenia fetida*), which perform best at temperatures between 55 and 80 degrees Fahrenheit. Once outside this range, particularly below 50 degrees, the microbial and metabolic rates of worms decline significantly (Angima et al., 2011). Flagstaff's winter temperatures, which are frequently below freezing and its higher elevation, result in cold, low-humidity conditions. The worms risk freezing to death and drying out. To compensate for this, systems must be housed indoors, in greenhouses, or in insulated structures. This requires utilities and dedicated campus space, both of which are often in short supply. The extra maintenance required for the area's environment makes vermicomposting a slightly more complex biological infrastructure system.

Small vermicomposting systems are inexpensive, but larger scaled operations require much more cost consideration. The most common type of large-scale vermicomposting operation uses continuous flow-through (CFT) systems. These models can get expensive, ranging from \$150-\$400 for in home units to \$1,700-\$5,000 for small commercial grade units (Old Tom's Wormery, n.d.). Large scale industrial units easily exceed the \$10,000-\$25,000+ range (Wormgear, 2026). Key factors influencing price include unit size, system volume, construction materials, and automated harvesting systems. Building a CFT system may be cheaper, but it would require sourcing materials and much manual labor. Monitoring equipment, such as industrial thermometers, moisture sensors, and data-collection equipment, will also need to be acquired. Bedding materials are low cost but need to be kept up, along with worms for restocking when populations decline.

Labor costs pose a primary constraint for a large scale project, as they are the most recurring cost. Workers are needed to regularly feed the systems, monitor moisture and pH levels, properly care for the worms to prevent odors or pests, and harvest worm castings. Since NAU relies primarily on student workers for research projects, a higher turnover rate requires more consistent supervision. Without stable funding to get a large scale operation up and running

and to employ full time staff, the system risks neglect, unnecessary complications, and loss of worms and productivity. The Green Fund is a good source for requesting startup costs, but it's not meant to be maintained long term.

Vermicompost requires clean and well sorted organic food waste to thrive. At NAU, this means coordinating with the dining halls to collect pre-consumer food waste. Since post-consumer food waste on campus contains meat, cheese, oils, and other harmful products, it is best to incorporate vermicompost into the current thermophilic composting system rather than replace it. This would mean the dining hall staff would need to separate their pre-consumer food waste into separate bins from the Somat machine. Facilities must support this effort and the transport of these materials, which require additional labor. This change is not just technical, but requires behavioral change, retraining of staff, and cross departmental collaboration. It also takes extra care when handling foodstock for vermicompost, as you do not want to overload your worm bins. Large scale operations would benefit from coordinating with thermophilic processes to predecompose some of the pre-consumer food waste to speed up the vermicomposting process. Dining halls generate waste continuously, which can pose a constraint for vermicomposting, which is a slower process. NAU may need multiple medium sized units closer to waste sources, or a dedicated large scale operations space.

B. Knowledge Gaps

A major contributor to knowledge gaps regarding vermicomposting capabilities on campus is the lack of data on the percentage of compostable food waste. The V.L.L has shown that coordination between dining halls can be done and supported, but on a large scale, it is unknown what a systematic change would look like. It would require frequent pickup and drop off of collection bins from the dining halls, and increase the workload of meal preppers. Data should be calculated to determine what percentage of NAU's food waste is suitable for worms and what percentage is contaminated. Seasonal variations in waste generation are a contributing factor, as students have varying high and low density semesters. Without this data, it is uncertain what size system is best for the campus, and feedstock may be inconsistent. An audit of waste generation and suitable substrates should be conducted before scaling.

The Vermicompost Living Lab is in the first year of production, with the first year of data expected to be analyzed in 2026. Without the data yet processed, there is insufficient annualized data from existing small systems, and they cannot appropriately be used for expansion. The smaller systems in place still serve as crucial preliminary information. To save time and money, observe these systems for pounds of waste processed per week, worm population trends, and seasonal performance changes. The existing research projects are a great starting point and should be utilized for further analysis before determining expansion goals. By analyzing the collected data these preliminary experiments serve as high quality research assets for the future of vermicomposting on campus.

In this feasibility study, cost comparisons of vermicomposting to thermophilic composting have been touched on. An NAU specific, indepth cost benefit analysis model would strengthen future expansion and grant proposals. Preliminary data have shown that castings provide a valuable return on investment for NAU in vermicomposting compared to composting. Castings are known as “black gold” to those in agriculture, due to their microbially rich nature and nutrient dense benefits. Institutional decisions require financial justifications and operational comparisons to ensure integral implementation. In depth cost models and their development could include the price per pound of compost relative to worm castings, the cost of student labor, and emissions reductions relative to yearly trends. The knowledge gaps in vermicomposting at NAU indicate that it can work on campus, but effective design requires steps to integrate it into an existing complex system.

VII. Conclusion & Next Steps

This feasibility study indicates that vermicomposting at NAU has strong potential to complement existing composting infrastructure on campus, particularly when its education, research, and campus sustainability initiatives are considered. Vermicomposting is not proposed as a replacement for thermophilic composting, but rather as an addition that expedites current operations, providing a more well-rounded and comprehensive approach to waste management. Successful scaling will require careful planning and research, including resource allocation and appropriate space, the development of efficient collection and transportation services, and coordination across campus departments. The V.L.L has already demonstrated feasibility on a medium sized scale and shown interest in collaboration among campus groups such as the Green Fund and the SSLUG Garden.

The next phase includes honing in on the efficiency of the V.L.L while proposing/creating new projects that specifically explore campus expansion. Key elements of this phase will include quantifying pre-consumer food collection on campus and using this data to determine appropriately sized system sizing and capacity requirements. Additional aspects include identifying suitable installation locations, refining waste collection and hauling methods, and evaluating associated costs.

A central concept associated with the V.L.L is intertwining research education with hands-on experience. The vermicomposting system will serve as a living laboratory, supporting undergraduate or graduate research while directly benefiting NAU. Vermicomposting opens the door to independent projects in sustainability, microbial community studies, environmental impacts, system design, and waste management. Ensuring that these systems remain open to further research and education is critical not only for the project's success but also for alignment with institutional goals.

Vermicomposting is a feasible, scalable system that offers NAU an opportunity to expand its waste management practices to be more sustainable. It is also advancing the university's commitment to further education, research, and student engagement. With a continued focus on expansion and sustainability, the V.L.L has the potential to expand into a campus wide operation, delivering both educational and environmental goals that further NAU's Climate Action Plan and commitment to carbon neutral waste management.

Next recommended steps:

1. Conduct a pilot expansion of the Vermicompost Living Lab using insulated, semi-automated systems.
2. Partner with dining to secure a small, uncontaminated, consistent waste stream.
3. Develop monitoring protocols for temperature, moisture, and output.
4. Collect and analyze 1–2 years of vermicomposting performance data.
5. Build a cost-benefit model comparing vermicomposting, existing Somat systems, and landfill disposal.
6. Seek Green Fund, grant, or research funding to support expansion

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